r08 — POWER DISTRIBUTION PCB / UNROUTED PLACEMENT / NOT FOR FABRICATION

Power distribution PCB circuit



File: Distribution.kicad_sch

Sheet: /
File: UnderBot.kicad_sch

**Title:**

| Size: A3 | Date: | **Rev:** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/2 |

# POWER DISTRIBUTION PCB | PLACEMENT DRAFT — NOT FOR FABRICATION

Input is NOCO 12V OUT via approved cable, EXTERNAL main fuse, service disconnect and qualified polarity protection.

J1
INPUT + / GND
VIN_PROTECTED 1
GND_SYS 2

J2
STOP LOOP: pin1 SEND+ / pin2 RETURN+ (NO GND)
VIN_PROTECTED 1
ACT_SWITCHED 2

J2 goes to off-board normally-open power contact.
External hardwired stop + manual reset must control it.
NO permanent bridge. This is NOT a logic input.
Both terminals are POSITIVE rails; neither is ground.

F1
MINI fuse rating HOLD
VIN_PROTECTED — PI_12

J3
PI MODULE + / GND
PI_12 1
GND_SYS 2

52Pi EP-0225 DC input -> supplied USB-C cable -> Pi5.

F2
MINI fuse rating HOLD
VIN_PROTECTED — DEPTH_12

J4
KINECT CONDITIONER + / GND
DEPTH_12 1
GND_SYS 2

12V regulation/adapter matching still requires qualification.

F3
MINI fuse rating HOLD
ACT_SWITCHED — ARM_12

J5
ARM VIN + / GND
ARM_12 1
GND_SYS 2

Documented Adeept VIN6-24V; verify revision + USB VBUS block.

F4
MINI fuse rating HOLD
ACT_SWITCHED — AUX_12

J6
AUX CONVERTER + / GND
AUX_12 1
GND_SYS 2

5V converter -> audio and camera pan/tilt. All aux stops with arm.

H1 M3 3.2mm
H2 M3 3.2mm
H3 M3 3.2mm
H4 M3 3.2mm

Converter modules are OFF-BOARD. Board distributes nominal12V; it does NOT directly provide5V or3.3V.

T12 has independent battery/stop. All fuse values, wire ratings, input protection and current limits remain release holds.

J pin numbers are proposed harness assignments, not OEM connector pinouts
PROVISIONAL CIRCUIT / HARNESS DESIGN — NOT FOR FABRICATION
**UnderBot**

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***Title: UnderBot — standalone fused distribution board r08***

| Size: A3 | Date: 2026-10-02 | **Rev: 0.8 PLACEMENT** |
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